

Scenario-Based Requirement Analysis
Smart Waste Collection and Route Optimization Platform

CSA10 / CSA1063 – SOFTWARE ENGINEERING

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Software Requirement Specification (SRS)

Introduction

The Smart Waste Collection and Route Optimization Platform is an IoT and AI-based smart city solution designed to monitor bin fill levels, optimize waste collection routes, and assist municipal waste management teams in efficient fleet operations in real time. The system provides real-time bin monitoring, dynamic route generation, predictive fill-level analytics, and assists city administrators in sanitation and fleet planning.

Problem Description

Urban waste collection inefficiency increases fuel consumption, operational cost, vehicle downtime, and environmental pollution. Existing waste collection systems rely on fixed, calendar-based schedules and lack real-time monitoring and predictive capabilities, resulting in overflowing bins in high-demand areas and wasted trips to near-empty bins. The proposed platform integrates IoT fill-level sensors, GPS tracking, and AI analytics to improve municipal waste collection efficiency and route planning.

Scope of the System

The platform supports:

- Real-time bin fill-level monitoring
- Route optimization and planning
- Dynamic route recalculation
- AI-based fill-level prediction
- Fleet tracking and analytics
- Multi-zone / multi-city deployment
- Driver mobile application support
- Web-based administration dashboard
- Citizen notification service
- Secure data management

Functional Requirements

- User Registration
- Authentication
- Truck / Vehicle Registration
- Bin Fill-Level Monitoring
- Route Generation
- Dynamic Route Recalculation
- Driver Navigation
- Collection Confirmation
- AI Prediction (Fill-Level Forecasting)
- Notifications
- Report Generation
- Dashboard Management
- Bin Issue Reporting
- Collection History
- Analytics

Non-Functional Requirements

- High Performance
- High Availability
- Reliability
- Scalability
- Security
- Maintainability
- Usability
- Portability
- Compatibility

Use Case Descriptions

UC1 Register User

Actor: Driver / Admin

Description: Creates a new account.

Precondition: User not registered.

Postcondition: Account successfully created.

UC2 Login

Actor: Driver / Dispatcher / Admin

Description: Authenticate user credentials.

Precondition: Registered account.

Postcondition: Dashboard displayed.

UC3 View Bin Status

Actor: Dispatcher

Description: Displays the real-time fill level of all bins on a live map.

UC4 Generate Optimized Route

Actor: Fleet Manager / System

Description: Generate the most efficient collection route based on flagged bins.

UC5 Predict Bin Fill Trends

Actor: AI Engine

Description: Predict future bin fill levels using historical and real-time sensor data.

UC6 Generate Analytics Report

Actor: Municipal Manager

Description: Generate reports for performance evaluation and city planning.

Data Requirements

Input Data

- Driver Details
- Truck / Vehicle Information
- Bin Fill-Level Data
- GPS Coordinates
- Sensor Data
- Route Data
- Collection History Data
- Citizen Subscription Data

External Interface Requirements

User Interface

- Driver Mobile App
- Dispatcher / Admin Web Dashboard
- Citizen Notification Portal

Hardware Interface

- IoT Fill-Level Sensors
- GPS Devices
- Truck-Mounted Devices
- RFID / Barcode Bin Scanners

Software Interface

- Route Optimization Engine (VRP-based)
- Mapping / Navigation API
- Notification Service
- Cloud Database

Communication Interface

- Wi-Fi
- 4G / 5G
- LTE
- MQTT
- REST API
- HTTPS

System Features

- Real-Time Bin Monitoring
- Live Bin Map
- AI-Based Fill Prediction
- Dynamic Route Optimization
- Real-Time Notifications
- Driver Navigation
- Collection Confirmation
- Fleet Tracking Dashboard
- Smart Analytics Dashboard
- Multi-Zone Management
- Secure Communication

Assumptions

- IoT sensors work correctly.
- GPS devices continuously provide accurate location.
- Internet connectivity is available across collection zones.
- Drivers use the mobile application.
- Bins are digitally connected and sensor-equipped.
- AI models are trained using sufficient historical data.
- Cloud servers remain operational.

Constraints

Technical Constraints

- Dependence on IoT devices
- GPS accuracy limitations
- Network latency
- Sensor battery life limitations

Operational Constraints

- Continuous sensor maintenance
- Staff training
- Real-time monitoring required

Economic Constraints

- High deployment cost
- Infrastructure maintenance cost
- Cloud hosting expenses

Legal Constraints

- User data privacy regulations
- Municipal compliance requirements

- Environmental regulation compliance

Time Constraints

- Limited project development timeline
- Periodic AI model updates
- Sensor installation schedule

Suggested Improvements

- Add emergency/priority collection routing for hazardous waste.
- Include recycling bin segregation tracking.
- Integrate smart traffic signal coordination for faster truck movement.
- Provide dynamic fuel-cost-based route budgeting.
- Support offline route caching for low-connectivity zones.
- Add multilingual mobile application support.
- Include predictive maintenance for IoT sensors and trucks.
- Implement carbon emission monitoring and reporting.
- Add disaster recovery and backup mechanisms.
- Integrate with municipal complaint management systems.